ALBATROSS — User Manual

Launch, mission planning and ground-control operation

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1 Start here

ALBATROSS is a browser ground-control station for PX4 simulation and connected real aircraft. Open http://localhost:8080 after launching the required stage. The map, mission editor, telemetry, aircraft 3D view and camera panes share this page.

For a first prototype session, use **Stage 1: planning simulation**. The October GUI changes passed offline checks; browser rendering, camera streaming and live mission execution still require verification on the host. This manual describes implemented behavior, not a completed flight test.

2 Docked workspace and floating controls

The workspace uses larger labels, telemetry values and circular controls. Choose **Large text** in the header for additional scaling; **Normal text** restores the default. The preference persists across reloads. Primary flight icons are consistent SVG shapes with text labels. Keyboard focus and reduced-motion preferences are supported.

Details includes a bounded, per-browser **Session events** log of flight requests, acknowledgements, confirmations and failures. A lost aircraft connection shows how long ago telemetry was last received and revokes backend mission verification. The launchers now check for conflicting real/simulation relay processes before restarting services. A blocked launch lists PIDs without stopping them. See `docs/SITL_ACCEPTANCE.md` for the repeatable host acceptance run.

The map is the main pane; one FPV pane and a labelled Gazebo 3D camera pane sit beside it. Gazebo is a streamed simulation camera, not an embedded interactive Gazebo application. In real mode that pane explains that simulation is unavailable. The **2D map**, **Aircraft 3D** and **GPS** tabs switch the main view.

Drag the shared dividers to resize neighboring panes. Focus a divider and use arrow keys for keyboard resizing. **Expand** maximizes a pane; Restore returns to the split layout. The minus button collapses a camera pane. **Reset layout** restores defaults. Split proportions are saved in the browser. Fullscreen applies to the complete workspace, keeping flight controls visible.

Mission opens the waypoint/boundary/settings drawer. **Details** opens detailed telemetry and altitude/speed controls. Close the drawer or press Escape to return to the views. The round flight controls float above the workspace; no permanent bottom control panel consumes viewing space.

Quick mission: click **Choose destination**, then click the 2D map. This replaces the local draft and plans a route automatically. Review the route, altitude and Auto-RTL settings in Mission. With a valid position and connection, press **ARM**, then wait for telemetry to confirm arming.

A separate **GO** press uploads and reads back the mission, verifies it again before starting, and requests mission mode. No GO action implicitly arms the aircraft.

HOVER requests PX4 hold without clearing the mission. The prototype requires a fresh global position/altitude and checks estimator absolute-position flags when available; local-position-only hover is not implemented. The flight controller decides whether the request is acceptable. **RETURN** requests RTL using the aircraft's settings. **RC manual** requests pilot control; **Resume auto** resumes the current mission item after readback verification. Real-mode RC takeover requires fresh received RC channels with nonzero reported RSSI (unknown RSSI is shown as unverified signal strength, not a measured quality). The RC indicator shows observed input, not a guarantee of receiver reliability. **Land here** is a separate confirmed action.

The red **STOP** circle opens the emergency motor-stop dialog. It requires explicit acknowledgement before sending forced disarm (not flight termination). An airborne aircraft can fall. The normal Disarm action remains in the Mission drawer. Emergency motor stop bypasses the normal command lock but still depends on a live communication link and aircraft response. Its physical behavior has not been flight-tested in this session.

Flight actions show pending/accepted state and wait for received telemetry before claiming the requested state. A timeout means the outcome is unknown; check the aircraft/QGC before retrying. Link loss disables the controls and invalidates the browser's mission verification. A QGC mission-ID change also invalidates it when the aircraft publishes supported IDs; readback before start/resume is always used.

Read aircraft mission / QGC changes in Mission downloads the stored mission and draws supported navigation items in amber. The blue local draft is unchanged. This is read-only inspection, not bidirectional draft synchronization. QGC must still download the aircraft mission after a web upload. QGC is no longer restarted automatically in either real or simulation mode. Use one mission writer at a time; web locking cannot prevent QGC from starting an independent transfer.

3 Choose and launch a stage

Run commands from the repository root. Simulation requires the project's PX4 SITL build/world, ROS 2 Humble and Gazebo installation. The launchers expect `external/PX4-Autopilot` and `external/mavlink2rest`; see the README for dependency setup. The browser loads map imagery and JavaScript libraries from internet services.

```
# Stage 1: autonomous planning simulation, no RC input
GCS_MODE=sitl bash launch/start.sh 1

# Stage 2: simulated aircraft flown using a real RC receiver
GCS_MODE=sitl bash launch/start.sh 2

# Stage 3: connected real aircraft, no Gazebo
bash launch/start.sh 3
```

Choose *one* stage. These launchers restart shared GCS/MAVLink services; changing stages interrupts the current connection. Run the scripts as files, not by pasting their contents into a terminal.

Stage 1 starts the web server, MAVLink bridge, PX4 SITL, Gazebo and chase/top-down cameras. Allow roughly 15 seconds for simulator and camera warmup; use live status rather than elapsed time to decide whether the connection is ready.

Stage 2 additionally reads RC channels from a USB-connected Pixhawk and forwards them to SITL. The receiver remains wired to the Pixhawk; the web GCS and QGroundControl control the simulator. If the USB device is missing, the launcher reports that the RC bridge did not start. Check `~/rc_bridge.log`.

Stage 3 starts the real-aircraft MAVLink relay, web GCS, front-camera bridge and QGroundControl. It does not apply SITL arming bypasses. The camera and telemetry links are independent. See `docs/HM30_LIVE_VIEW.md` for connection checks and `docs/QGC_SETUP.md` for QGroundControl setup.

In the Mission drawer, the in-page **Plan**, **Manual sim** and **Launch SITL** buttons launch simulation processes in a SITL session. They do not switch a real-aircraft server into simulation mode. Simulator controls are hidden in real mode and the server rejects simulator start/stop/follow requests there. **Digital twin** (or **Aircraft 3D** in real mode) opens a visualization; flight controls still act on the connected aircraft.

4 Read the connection and displays

- **LINK** means a recent aircraft heartbeat is available. **NO LINK** means current flight state is unavailable, even if video plays.
- A heartbeat alone does not provide position. Wait for usable GPS before planning a real mission. The position banner distinguishes missing telemetry from a connected aircraft waiting for GPS. The 3D model hides without position; a faded 2D marker can remain at the last received location.
- Dashboard values show received speed, relative altitude, heading, mode and battery data. Dashes indicate unavailable values. Heading is aircraft orientation, not direction of travel; missing attitude data dims the horizon.
- **Locate** centers/follows the aircraft when position is available. Before GPS, browser location permission can center the map on the operator. Dragging the map disables following; click Locate to resume.
- **Aircraft 3D** opens the aircraft view; **2D map** returns to the map. The 3D **Follow** button tracks the model. Route edits/clears update both views. If the 3D library is unavailable, the GUI returns to 2D and reports it.
- **GPS view** includes an illustrative satellite constellation. Its satellite positions and inferred locks are not measured orbital data; use the received GPS fix, satellite count and accuracy readouts for status.

5 Set the operating boundary

The initial boundary is the saved polygon, or the default KML boundary if none has been saved. Exclusion zones are drawn in red.

1. Click **Draw new** and draw a polygon (double-click to finish), or **Edit / drag** to change the existing boundary.
2. Click **Save**. Wait for the "Boundary saved" message.
3. To restore the default KML boundary, click **Reset**, then Save.

Any boundary edit or reset clears the planned route. Planning remains blocked until the edited boundary is saved. A failed save leaves the edits unsaved; correct the reported error and retry. If the boundary changes during saving, save again. Saving a boundary does not alter a mission already uploaded to the aircraft.

6 Plan and upload a mission

1. Check the operating boundary and connection. In real mode, wait for live aircraft GPS and reported home position.
2. Set **Altitude**, click **Add waypoints**, then click the map in the intended order. Click Add waypoints again to leave placement mode.
3. Edit individual heights in the waypoint list; the remove control deletes a waypoint. The waypoint **Clear** button removes the local draft. These edits clear the preview, not the aircraft's uploaded mission.
4. Click **Plan route (avoids no-fly)**. The planner routes around exclusions within the saved boundary. A blocked target may be moved to a safe edge; review the warning and resulting endpoint before uploading.
5. Set **Auto-RTL @** and its return altitude as required. With this option enabled, the uploaded mission includes a return-altitude waypoint at the final outbound point, a planned return path and a landing item. In real mode the destination is the aircraft-reported home, not the map's default base. If home is unavailable, the upload fails with an explanatory message.
6. Click **Upload & verify** and wait for success. Upload waits for a new addressed aircraft mission acknowledgement and verifies stored items by readback. Invalid routes or routes crossing the current boundary/exclusions are rejected; plan again if the boundary changed.
7. Click the floating **ARM**, verify the received armed state, then **GO**. In real mode Arm asks for confirmation, and GO requires the aircraft to be armed separately. Confirm the resulting mode in live telemetry.

The default configured altitude ceiling is 60.96 m; whole-metre GUI editors use 60 m. The minimum waypoint altitude is 5 m and the return-altitude input minimum is 10 m. Server configuration can change the ceiling. This software limit is not an assessment of permission to fly at a location.

Replan after changing waypoint positions/heights or the boundary. An old route is cleared before a new planning request; late responses cannot restore a route after further edits. Plan and Upload disable while their requests are pending.

Preview limitation: the map/3D preview shows the outbound route only. Return segments are generated and checked at upload. With Auto-RTL unchecked, the uploader adds no automatic return/landing sequence; final behavior depends on the flight controller. The separate **RTL** button

requests PX4's RTL mode and uses its configured return behavior, not the mission editor's explicit return-leg construction.

7 Flight controls

Control	Behavior
ARM / Disarm	Sends the corresponding request. Normal Disarm is in Mission. The label follows received armed state.
GO	Uploads if needed, reads back the aircraft mission, then starts from its first item. Requires separate arming in both modes.
Resume auto	Verifies the aircraft mission and requests mission mode from the current item. It does not restart from item zero.
RC manual / Resume auto	RC manual requests pilot position control (POSCTL by default); Resume auto continues the verified mission. This is separate from the Manual sim launcher.
Target spd	Changing the input/slider sends a speed request after a short pause (about 0.5 s). No Alt Hold click is required for speed.
Altitude / Alt Hold	Adjusting altitude sets the target for new waypoints. Alt Hold applies it at the current received position and requests a hold. It requires live GPS.
Fly to last WP	Requests a direct reposition to the final draft waypoint and its height. Requires live GPS; blocked destinations or direct paths are rejected. Use Plan route for a detour.
RETURN	Requests the flight controller's return-to-launch mode. Verify the mode change in telemetry.
Stop / Clear mission	Requests hold/loiter and clears the aircraft mission. On successful delivery it clears the local route preview. It is not a land or disarm command. Upload again before starting a replacement mission.
Drop Cargo	After confirmation, sends the configured payload servo request. Actual release depends on aircraft actuator setup.

Flight commands wait for matching aircraft acknowledgements. Acceptance is not confirmed execution; floating controls additionally wait for the requested telemetry state. Check live mode, armed state and motion. Mission-upload acknowledgement is distinct from flight-command acknowledgement. If a command fails, read the mission status message before retrying.

8 Camera operation

In SITL, click **start** in the **CHASE CAM** or **TOP-DOWN CAM** pane. The second pane's **FPV / Top-Down** button changes its simulated view. The feeds can initially show a waiting image while Gazebo starts.

In real mode, the **FRONT CAMERA** pane starts automatically. Its status shows live, waiting, paused or bridge unavailable. Video status does not establish an aircraft telemetry connection. Use **stop/start** to pause/resume the pane, **Enlarge**, fullscreen or the resize corner to inspect the video, and **Reset** to restore the camera size. This Reset is separate from boundary Reset.

9 Troubleshooting

Message or symptom	Next step
Operating mode unavailable	Check the web server, then reload. The GUI does not assume simulation for arming when mode discovery fails.
No live heartbeat / NO LINK	Check the selected stage and MAVLink bridge/relay. Camera video alone is insufficient.
Waiting for GPS / reposition rejected	Wait for valid received aircraft position. Operator geolocation and the map center are not aircraft GPS.
Save edited boundary	Save successfully, then plan a new route. Boundary editing invalidates the previous preview.
Home position unavailable	Wait for aircraft HOME_POSITION before an explicit real-aircraft return-home upload. Browser coordinates cannot substitute for it.
Route crosses boundary / exclusion	Check the saved boundary and waypoints, then replan. Uploaded segments are checked again against current geometry.
3D unavailable / blank map	Check the internet connection for CDN scripts, imagery and terrain; use 2D if 3D is unavailable.
Camera waiting / bridge unavailable	Check the camera bridge and source independently of telemetry. See the HM30 guide for real-camera network checks.
GO reports disarmed	Arm separately and verify the armed state before GO.

Read-only service checks:

```
curl --max-time 3 http://localhost:8080/api/mode
curl --max-time 3 http://localhost:8080/api/telemetry
curl --max-time 3 http://localhost:8091/healthz
# Real-mode front camera health through the GCS:
curl --max-time 3 http://localhost:8080/api/camera/status
```

Useful logs are `~/gcs_server.log`, `~/sitl_clean.log`, `~/chase_cam.log` and `~/rc_bridge.log` for simulation, plus `~/mav_hub.log`, `~/mavlink2rest.log` and `~/rtsp_bridge.log` for real hardware.

10 Reload, restart and stop

Reload the browser after GUI changes. Python changes require restarting the GCS process. For a SITL session, the existing helper restarts the web server and chase camera while leaving PX4/Gazebo running:

```
GCS_MODE=sitl bash launch/restart_gcs.sh
```

This helper starts a Gazebo chase camera and is not the real-hardware restart workflow. For a grounded real-hardware session, rerun Stage 3 with the same connection settings; it also restarts the relay, video bridge and QGroundControl.

Stop SITL terminates simulator, MAVLink bridge and simulation-camera processes; it leaves the web server running. It does not replace the aircraft **Stop / Clear mission** control. In Stage 2, the RC bridge is a separate process and is not stopped by the Stop SITL endpoint.

11 Prototype checks and limitations

The offline checks exercise UI bindings, route editing, error handling, mission construction and mocked aircraft responses:

```
python3 -m unittest discover -s gcs/tests -v
node gcs/tests/test_ui_commands.cjs
node gcs/tests/test_workspace.cjs
```

As of this update, 55 Python and 27 JavaScript tests pass. These do not establish physical link reliability, browser rendering, camera operation or flight behavior. The planner uses sampled/grid geometry; it is not a formally verified flight-safety system. QGroundControl must download the aircraft mission after web upload to inspect it; the two draft editors are not synchronized.

Ports at a glance

`8080`	GCS web UI, API and video proxy routes
`8088`	mavlink2rest REST API
`8091`	SITL chase camera or real front-camera MJPEG bridge
`8092`	SITL second camera (top-down / FPV)
`14555`	MAVLink to web bridge (UDP)
`14550`	MAVLink to QGroundControl (UDP)